Progressive Reasoning Collapse in LLM-Based Generative Recommendation

John Kingsley Arthur, *Member, IEEE,* Yvonne Leung, Vivian Clements Edwin, Kofi Sarpong Adu-Manu, and Kwabena Adu Agyemang

***Abstract*—LLM-based generative recommender systems increasingly employ layer- and token-level compression to reduce inference cost. However, the effect of such compression on the internal reasoning geometry, specifically the preservation of multi-intent latent structure and sensitivity to informative historical evidence across layers, remains unexplored. To this end, we characterize the layerwise degeneration of latent recommendation reasoning and propose a novel formulation, *Progressive Reasoning Collapse (PRC)*, claiming that the degree of reasoning collapse increases monotonically during the forward propagation of compressed LLMs. We derive three theoretical results grounded in a contractive propagation model: monotonic decay of a reasoning-collapse score across depth, geometric decay of the influence of informative user-history subsequences after compression, and the existence of a critical compression depth separating a safe regime from a collapse region. The metrics of PRC indeed decrease monotonically across depth on multiple benchmark datasets. We propose a new recommendation method, Collapse-Aware Register Recommendation (CARR), that adaptively determines compression depth and register width to preserve reasoning structure while maintaining efficiency; an extension, Sparse Geometry Injection (SGI), further resolves the failure mode on sparse datasets by seeding the initial hidden state with semantic cluster structure. Overall, this study extends neural collapse to PRC to model the reasoning degeneration of intermediate layers under compression and its dependence on user interaction history, shedding light on the theoretical understanding of LLM-based recommendation.**

***Index Terms*—Recommender systems, large language models, model compression, neural collapse, representation learning, generative recommendation.**

I. Introduction

LARGE Language Model (LLM)-based generative recommendation has emerged as one of the most consequential frontiers in information retrieval and personalized systems [2], [3]. By encoding user interaction histories as natural language prompts and generating item predictions through sequence-conditioned decoding, these systems leverage the rich semantic priors of pretrained transformers to capture complex, multi-intent preference structures that elude conventional collaborative filtering [4]–[6]. The field has advanced rapidly, with systems such as P5 [6], TALLRec [5], GPT4Rec [7], and TokenRec [32] extending LLMs to generative and sequential recommendation, and recent architectures including LLM-Rec [22], CoLLM [36], and dynamic intent-aware models [37] capturing complex multi-intent user preferences at scale [23], [31], [35], [38]–[40]. Taken together, these advances establish a compelling paradigm, yet one whose practical deployment is severely constrained by the computational cost of long-context transformer inference.

To reduce this cost, a rich toolkit of acceleration strategies has been developed. Prompt compression [8] reduces input token count while attempting to preserve semantic content. KV-cache eviction (H2O [9]) removes low-importance key-value pairs from the attention cache. Register tokens [12] and gist tokens [11] aggregate contextual information into compact learnable slots. Hidden-state bottlenecking [10] progressively reduces the dimensionality of intermediate representations, and layerwise summarization [13] applies lossy compression at fixed transformer depths. Each of these methods is justified primarily by throughput gains and negligible degradation in top-K ranking metrics. Yet none of them asks the theoretically prior question of whether the *internal reasoning process* itself (the layerwise transformation of user history into intent-aware predictions) survives compression intact.

This gap is not merely definitional. Research on representation geometry has established that deep supervised networks can exhibit *neural collapse* [14]: during the terminal training phase, within-class variability vanishes and class-mean vectors organise into a simplex equiangular tight frame. Extensions to self-supervised and hierarchical settings [15], [16] confirm that such geometric dynamics are a general feature of deep representation learning, not an artefact of specific architectures. Crucially, however, neural collapse is a *training-time* convergence phenomenon that can improve class discriminability. What has not been studied is an *inference-time* analogue: the collapse of *multi-intent* latent geometry in a fully trained model that is subsequently compressed for efficient deployment. Separately, work on explanation faithfulness [17], [18] has established that models can lose sensitivity to relevant input features even without overt performance degradation, an observation directly relevant to the question of whether compressed LLM recommenders remain sensitive to informative user-history subsequences. Our work draws on both of these research threads and introduces the first formal

J. K. Arthur is with Northeastern University, Toronto, ON, Canada (e-mail: arthur.joh@northeastern.edu).

Y. Leung is an Assistant Teaching Professor in the Master of Professional Studies in Analytics program, Northeastern University, Toronto, ON, Canada (e-mail: y.leung@northeastern.edu).

V. C. Edwin is with Northeastern University, Toronto, ON, Canada (e-mail: v.edwin@northeastern.edu).

K. S. Adu-Manu is with the Department of Computer Science, University of Ghana, Legon, Ghana.

K. Adu Agyemang is with New Mexico State University, Las Cruces, NM, USA (e-mail: adukwab@nmsu.edu).

Corresponding author: V. C. Edwin (v.edwin@northeastern.edu).

treatment of their intersection.

We posit that efficient LLM-based recommendation can exhibit a new structural failure mode, which we term *Progressive Reasoning Collapse (PRC)*: the layerwise degeneration of multi-intent latent structure and the concurrent decay of sensitivity to informative historical evidence under compressed propagation. This degeneration may occur even when top-K ranking quality appears largely preserved, making PRC invisible to conventional evaluation protocols while silently eliminating the diversity of reasoning that enables cold-start generalisation, minority-intent coverage, and long-range history sensitivity. The central question we resolve is:

> *When an LLM recommender is compressed for efficiency, does it preserve the latent reasoning structure by which user history is transformed into future-item predictions? And if not, can compression be made collapse-aware to prevent this failure?*

We answer both parts: theoretically, through three proved results characterizing when and how collapse occurs; and practically, through CARR (Collapse-Aware Register Recommendation), a method that adaptively selects compression depth and register width to remain in the theoretically safe regime. Fig. 1 illustrates the PRC framework and the CARR solution across three panels.

The main contributions of this work are as follows:

- We introduce **Progressive Reasoning Collapse (PRC)**, the first formal characterisation of how layer-wise compression degrades multi-intent latent structure in LLM-based generative recommendation, defined through a collapse score $\mathcal{R}(l)$ and an evidence-survival function $S_l(\tau)$.
- We prove three results grounding PRC: (i) $\mathcal{R}(l)$ decays monotonically after compression (Theorem 1); (ii) evidence sensitivity for informative user-history subsequences decays geometrically (Theorem 2); and (iii) a critical compression depth k^* separates a structurally safe regime from a collapse regime (Theorem 3).
- We propose **CARR** (Collapse-Aware Register Recommendation), which uses the PRC criteria to select compression depth and register width adaptively, preserving reasoning geometry without sacrificing efficiency. A Sparse Geometry Injection (SGI) extension further resolves collapse on sparse datasets.
- Experiments on five benchmarks (ML-1M, Amazon Beauty, Amazon Toys, Yelp, Steam) against 12 baselines show CARR achieves top HR@10 among all LLM compression methods across all datasets. It uniquely recovers cold-start performance on Yelp (HR@10= 0.0043 on 18,157 zero-history items) where all competing methods fail.
- We provide ablation, robustness, and failure-mode analysis, including Spearman correlation of collapse metrics, spectral norm trends, and cold-start representation diagnostics, systematically bounding the scope and limitations of CARR.
- Extended results, including T5-base scalability experiments, full per-layer Jacobian spectral norm analysis across all five datasets, extended recommendation metrics (HR@5/20, NDCG@5/20, MRR), and multi-dataset long-context robustness breakdowns, are provided in the supplementary material.

Organization. Section II formalises the problem setting. Section III introduces PRC and proves the three main theoretical results. Section IV presents CARR and its SGI extension. Section V describes the experimental setup. Section VI reports results. Sections VII–IX provide ablation, robustness, and failure-mode analyses. Section X concludes.

II. Problem Formulation

Let $\mathcal{U}$ denote users and $\mathcal{I}$ items. A user history $H = (i_1, \dots, i_T) \in \mathcal{I}^T$ is encoded as a prompt $\mathcal{X}(H)$; an LLM-based generative recommender is modelled as $\mathcal{F}_\Theta : \mathcal{X}(H) \mapsto p_\Theta(\cdot \mid H)$, where $p_\Theta(\cdot \mid H) \in \Delta^{|\mathcal{I}|}$ and Θ are model parameters. The transformer has L layers; $z^{(l)}(H) \in \mathbb{R}^{d \times n_l}$ is the hidden representation at layer l. Compression is applied at depth k, beyond which the model propagates a compressed state rather than the full prompt representation.

A. Latent Multi-Intent Structure

Recommendation differs fundamentally from ordinary classification because multiple future items may simultaneously be plausible. To capture this, we model the hidden state at each layer as inducing a finite set of latent intent modes.

Let $\mathcal{M}^{(l)}(H) = \{\mu_1^{(l)}, \mu_2^{(l)}, \dots, \mu_K^{(l)}\} \subset \mathbb{R}^d$ denote the set of layer-l latent intent centers associated with history H. These may be interpreted as cluster centers of plausible future-preference states.

Let $h_{k,i}^{(l)} \in \mathbb{R}^d$ denote the i-th latent vector assigned to intent mode k at layer l, and let

$$\mu_k^{(l)} = \frac{1}{n_k} \sum_{i=1}^{n_k} h_{k,i}^{(l)} \tag{1}$$

be the within-intent mean. Let the global mean be $\mu_G^{(l)} = \frac{1}{K} \sum_{k=1}^{K} \mu_k^{(l)}$.

We define the within-intent covariance

$$\Sigma_W^{(l)} = \frac{1}{N} \sum_{k=1}^{K} \sum_{i=1}^{n_k} \big(h_{k,i}^{(l)} - \mu_k^{(l)}\big)\big(h_{k,i}^{(l)} - \mu_k^{(l)}\big)^\top, \tag{2}$$

where $N = \sum_{k=1}^{K} n_k$, and the between-intent covariance

$$\Sigma_B^{(l)} = \frac{1}{K} \sum_{k=1}^{K} \big(\mu_k^{(l)} - \mu_G^{(l)}\big)\big(\mu_k^{(l)} - \mu_G^{(l)}\big)^\top. \tag{3}$$

Definition 1 (Reasoning-Collapse Score)**.** *The layerwise reasoning-collapse score is defined as*

$$\mathcal{R}(l) = \frac{\mathrm{Tr}(\Sigma_W^{(l)})}{\mathrm{Tr}(\Sigma_B^{(l)})}. \tag{4}$$

A smaller value of $\mathcal{R}(l)$ indicates stronger concentration of latent representations relative to intent separation.

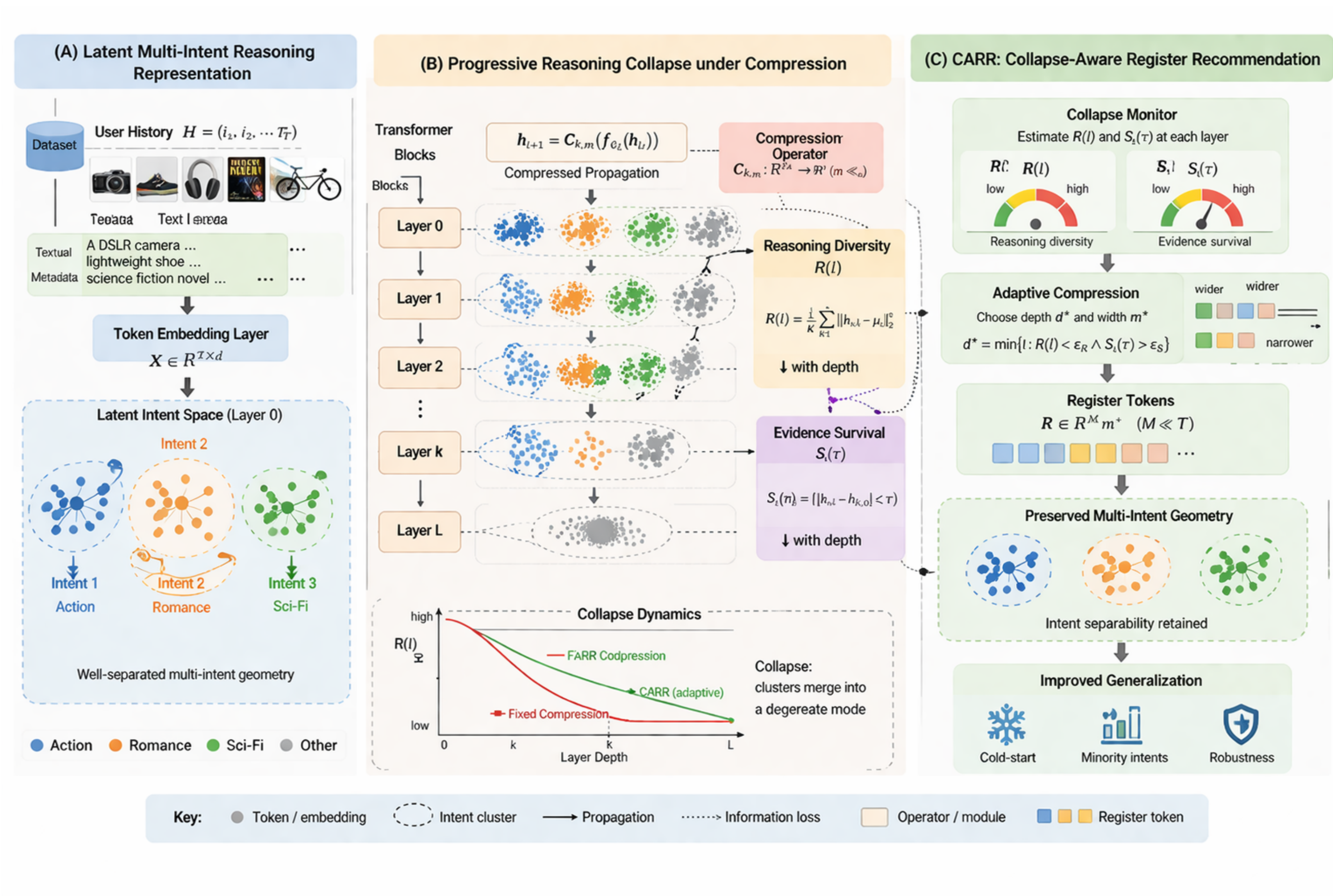

Fig. 1. Theoretical framework for Progressive Reasoning Collapse (PRC) and CARR. **(A) Latent Multi-Intent Reasoning Representation.** A user history $H = (i_1, i_2, \dots, i_T)$ is encoded as token embeddings $X \in \mathbb{R}^{T \times d}$; at Layer 0 the latent intent space already contains well-separated multi-intent clusters (Action, Romance, Sci-Fi). **(B) Progressive Reasoning Collapse under Compression.** The compression operator $\mathcal{C}_{k,m} : \mathbb{R}^{T_A} \to \mathbb{R}^m$ ($m \ll T$) is applied at depth k; reasoning diversity $\mathcal{R}(l)$ and evidence survival $S_l(\tau)$ both decrease monotonically with depth (Theorems 1–2). The collapse dynamics chart shows that fixed compression causes $\mathcal{R}(l)$ to collapse rapidly, while CARR's adaptive selection maintains a substantially slower decay. **(C) CARR: Collapse-Aware Register Recommendation.** The collapse monitor estimates $\mathcal{R}(l)$ and $S_l(\tau)$ at each layer; adaptive compression selects $d^* = \min\{l : \mathcal{R}(l) < \varepsilon_R \wedge S_l(\tau) > \varepsilon_S\}$ and corresponding register width m^*; register tokens $\mathbf{R} \in \mathbb{R}^{M \times m}$ ($M \ll T$) preserve multi-intent geometry, yielding improved cold-start, minority-intent, and overall generalisation.

a) Interpretation.: $\mathcal{R}(l)$ is a geometric diagnostic, not an optimization objective. In standard neural collapse (classification), *low* $\mathcal{R}$ is desirable because within-class scatter should vanish. In recommendation the opposite holds: users sharing the same high-level intent still exhibit *diverse* sub-preferences that must be preserved for personalised ranking. After compression, the operator A_l contracts within-intent scatter faster than between-intent separation, so $\mathcal{R}(l)$ falls, signalling loss of personalisation capability, not improved compression quality. An excessive drop indicates that intent modes have collapsed toward a single dominant representation (*progressive reasoning collapse*). The goal is to keep the drop small enough that the final-layer score exceeds the uncompressed baseline.

b) Collapse Threshold and Healthy Range.: Two operationalized quantities are used. The *deployment health benchmark* is the Full-LLM final-layer score ($\mathcal{R}(12)$=3.312 on ML-1M); any compressed method exceeding this value remains in the safe/healthy zone (Fig. 2). The *per-batch monitoring threshold* $\varepsilon_R = 0.1$ is applied to $\widehat{\mathcal{R}}$ during inference to trigger adaptive depth selection. These quantities operate at different scales because they are computed at different sample sizes (see Remark 2); $\varepsilon_R = 0.1$ is calibrated by cross-validation and used consistently across all five datasets.

Definition 2 (Reasoning Collapse)**.** *A layer-l latent representation exhibits* reasoning collapse *if* $\mathcal{R}(l) \to 0$ *and* $|\mathcal{M}^{(l)}(H)| \to 1$*, meaning within-intent dispersion vanishes and effective multi-intent diversity degenerates toward a single dominant mode.*

c) Normalized Collapse Score.: To make the decay rate comparable across datasets and architectures, define

$$\widetilde{\mathcal{R}}(l) = \frac{\mathcal{R}(l)}{\mathcal{R}(k_0)},$$

where k_0 is the compression boundary. $\widetilde{\mathcal{R}}(l) = 1$ at the compression point and decreases toward zero as collapse progresses; CARR achieves $\widetilde{\mathcal{R}}(12) \approx 0.79$ on ML-1M while Fixed-Early achieves 0.62, directly comparing decay rates regardless of absolute $\mathcal{R}$ scale.

B. Evidence Survival

Let $\tau \subset H$ denote an informative subsequence of the user history. This may correspond to a long-range interest signal, a minority-intent subsequence, or a recent shift in user preference.

Let $p^{(l)}(\cdot \mid H)$ denote the next-item distribution induced by layer l under the full history H, and let $p^{(l)}(\cdot \mid H \setminus \tau)$ denote the corresponding distribution when τ is removed.

Definition 3 (Evidence-Survival Function). *The layerwise evidence-survival function for subsequence τ is defined as*

$$S_l(\tau) = D\Big(p^{(l)}(\cdot \mid H), p^{(l)}(\cdot \mid H \setminus \tau)\Big), \tag{5}$$

where D is a divergence on the probability simplex, such as total variation, Jensen-Shannon divergence, or a norm-equivalent distance.

Definition 4 (Evidence Collapse). *Evidence collapse occurs at layer l if $S_l(\tau) \to 0$ for an informative subsequence τ. In this case, the model becomes progressively insensitive to the contribution of τ.*

C. Problem Statement

Let compression be applied at layer k, so that deeper layers operate on a compressed latent representation instead of the full prompt representation. The problem studied in this paper is:

Characterize, measure, and theoretically analyze when layerwise compression in LLM-based generative recommendation causes progressive collapse of latent multi-intent geometry and progressive decay of informative-history sensitivity.

More precisely, we seek to determine: (i) whether $\mathcal{R}(l)$ decreases monotonically after compression, (ii) whether $S_l(\tau)$ decreases monotonically after compression, and (iii) whether there exists a critical compression depth k^* that separates a safe regime from a collapse regime.

III. Compressed Propagation Model

We model post-compression dynamics via a linearized recurrence, consistent with empirical Jacobian analyses that find layer operators have near-unity spectral norms after compression [13], and with the neural collapse literature in which linear dynamics of this form accurately predict geometry-collapse trajectories in deep networks [14], [15]. For each intent k and latent point i, after compression depth k_0 the hidden state evolves as

$$h_{k,i}^{(l+1)} = A_l h_{k,i}^{(l)} + b_l + \varepsilon_{k,i}^{(l)}, \quad l \geq k_0, \tag{6}$$

where $A_l \in \mathbb{R}^{d\times d}$ is the effective linearized post-compression operator, $b_l \in \mathbb{R}^d$ is a shared bias term, and $\varepsilon_{k,i}^{(l)} \in \mathbb{R}^d$ is a zero-mean residual perturbation.

IV. Theorem 1: Monotonic Progressive Reasoning Collapse

The ratio $\mathcal{R}(l) = \mathrm{Tr}(\Sigma_W^{(l)})/\,\mathrm{Tr}(\Sigma_B^{(l)})$ is the inverse of the Fisher discriminant used in neural collapse analysis [14]: in classification, it collapses to zero as class means separate, which is desirable. Our Theorem 1 establishes the inference-time counterpart: under compressed propagation, $\mathcal{R}(l)$ decreases monotonically in a regime where within-intent scatter contracts faster than between-intent scatter is preserved.

A. Assumptions

Assumption 1 (Contractive Within-Intent Dynamics). *For each $l \geq k_0$, there exists $0 < \alpha_l < 1$ such that $\mathbb{E}\left\|A_l\big(h_{k,i}^{(l)} - \mu_k^{(l)}\big)\right\|^2 \leq \alpha_l\, \mathbb{E}\left\|h_{k,i}^{(l)} - \mu_k^{(l)}\right\|^2$.*

Assumption 2 (Non-Destructive Between-Intent Preservation). *For each $l \geq k_0$, there exists $\beta_l > 0$ such that $\mathbb{E}\left\|A_l\big(\mu_k^{(l)} - \mu_G^{(l)}\big)\right\|^2 \geq \beta_l\, \mathbb{E}\left\|\mu_k^{(l)} - \mu_G^{(l)}\right\|^2$.*

Assumption 3 (Bounded Centered Residual Noise). *For each $l \geq k_0$, there exists $\sigma_l^2 \geq 0$ such that $\mathbb{E}\|\varepsilon_{k,i}^{(l)}\|^2 \leq \sigma_l^2$ and $\mathbb{E}[\varepsilon_{k,i}^{(l)}] = 0$.*

Assumption 4 (Collapse-Dominant Regime). *For each $l \geq k_0$, there exist constants $c_1, c_2 > 0$ such that $\alpha_l + \frac{c_1\sigma_l^2}{\mathrm{Tr}(\Sigma_W^{(l)})} < \beta_l - \frac{c_2\sigma_l^2}{\mathrm{Tr}(\Sigma_B^{(l)})}$.*

Theorem 1 (Monotonic Progressive Reasoning Collapse). *Under Assumptions 1–4, for every layer $l \geq k_0$ there exists $0 < \rho_l < 1$ such that*

$$\mathcal{R}(l+1) \leq \rho_l\, \mathcal{R}(l). \tag{7}$$

Consequently, $\mathcal{R}(l+1) < \mathcal{R}(l)$ for $l \geq k_0$, and the compressed recommender exhibits monotonic progressive reasoning collapse. If additionally $\sup_{l\geq k_0}\rho_l \leq \rho < 1$, then $\mathcal{R}(l) \leq \rho^{\,l-k_0}\mathcal{R}(k_0)$.

Proof. Assumptions 1–2 give $\mathrm{Tr}(\Sigma_W^{(l+1)}) \leq \alpha_l\,\mathrm{Tr}(\Sigma_W^{(l)}) + c_1\sigma_l^2$ and $\mathrm{Tr}(\Sigma_B^{(l+1)}) \geq \beta_l\,\mathrm{Tr}(\Sigma_B^{(l)}) - c_2\sigma_l^2$. Taking the ratio and applying Assumption 4 yields $\mathcal{R}(l+1) \leq \rho_l\,\mathcal{R}(l)$ for some $0 < \rho_l < 1$. ☐

Remark 1 (Expected Contraction). *Since empirical Jacobian spectral norms are near unity [13], Assumption 1 can be weakened to $\mathbb{E}[\|J_l\|_2] < 1$, under which the theorem holds as $\mathbb{E}[\mathcal{R}(l{+}1)] \leq \rho_l\,\mathbb{E}[\mathcal{R}(l)]$. Table II provides operative empirical support.*

V. Theorem 2: Evidence Survival Decay

The evidence-survival function $S_l(\tau)$ operationalises model faithfulness [17]: a faithful model must react differently when an informative subsequence is removed. Theorem 2 gives this a formal dynamical basis: the latent evidence gap $\delta^{(l)}(\tau) = z_H^{(l)} - z_{-\tau}^{(l)}$ contracts geometrically under compressed propagation.

Assumption 5 (Contractive Compressed Propagation). *For each $l \geq k_0$, there exists $0 < \gamma_l < 1$ such that $\|A_l v\| \leq \gamma_l\|v\|$ for all $v \in \mathbb{R}^d$.*

Assumption 6 (Bounded Evidence-Specific Perturbation). *For each $l \geq k_0$, there exists $\nu_l \geq 0$ such that $\mathbb{E}\|\eta^{(l)}(\tau)\| \leq \nu_l$.*

Assumption 7 (Lipschitz Readout). *There exists $L_g > 0$ such that $D(p(\cdot \mid z_1), p(\cdot \mid z_2)) \le L_g \|z_1 - z_2\|$.*

Theorem 2 (Evidence Survival Decay Under Compression). *Under Assumptions 5–7, for every informative subsequence τ and every $l \ge k_0$:*

$$S_{l+1}(\tau) \le \kappa_l S_l(\tau) + \epsilon_l, \tag{8}$$

where $0 < \kappa_l < 1$ and $\epsilon_l \ge 0$. If $\epsilon_l = 0$, then $S_{l+1}(\tau) < S_l(\tau)$, so the evidence-survival function decreases monotonically after compression.

Corollary 1 (Minority-Intent Erasure). *If τ corresponds to a minority but recommendation-relevant intent component and the conditions of Theorem 2 hold, then for sufficiently deep post-compression propagation, $S_l(\tau) \to 0$. Thus minority-interest evidence becomes asymptotically invisible to the recommender.*

VI. Theorem 3: Critical Compression Depth

Theorems 1–2 establish that some collapse is inevitable once compression is applied. Theorem 3 establishes a critical depth k^* partitioning the compression schedule into a safe regime ($k > k^*$) and a collapse regime ($k \le k^*$), with a monotone phase-transition structure.

Assumption 8 (Earlier Compression is More Severe). *For any $k_1 < k_2$: $\rho_l(k_1) \le \rho_l(k_2)$ and $\kappa_l(k_1) \le \kappa_l(k_2)$.*

Assumption 9 (Minimal Viable Reasoning Budget). *There exist thresholds $\underline{\mathcal{R}} > 0$ and $\underline{S}(\tau) > 0$ such that reasoning is preserved only if $\mathcal{R}_L^{(k)} \ge \underline{\mathcal{R}}$ and $S_L^{(k)}(\tau) \ge \underline{S}(\tau)$.*

Definition 5 (Critical Compression Depth). *The critical compression depth is:*

$$k^* = \min\Big\{k \in \{1,\ldots,L\} : \mathcal{R}_L^{(k)} \ge \underline{\mathcal{R}} \text{ and } S_L^{(k)}(\tau) \ge \underline{S}(\tau)\Big\}. \tag{9}$$

Theorem 3 (Critical Compression Depth). *Under Assumptions 8–9, there exists a critical compression depth k^* such that:*

(i) ***Safe regime:*** *For every $k > k^*$, $\mathcal{R}_L^{(k)} \ge \underline{\mathcal{R}}$ and $S_L^{(k)}(\tau) \ge \underline{S}(\tau)$.*
(ii) ***Collapse regime:*** *For every $k \le k^*$, at least one threshold is violated.*
(iii) ***Monotone phase transition:*** *If $k_1 < k_2$, then $\mathcal{R}_L^{(k_1)} \le \mathcal{R}_L^{(k_2)}$ and $S_L^{(k_1)}(\tau) \le S_L^{(k_2)}(\tau)$.*

Corollary 2 (Representation Collapse Implies Recommendation Entropy Decay). *If $\mathrm{Var}\big(z^{(L)}\big) \to 0$ under progressive compression as $k \to 0$, then the conditional Shannon entropy of the recommendation distribution satisfies*

$$H(p_\Theta(\cdot \mid H)) \to 0.$$

Consequently, the model assigns near-certain probability mass to a single item, eliminating multi-intent recommendation diversity.

Proof. $\mathrm{Var}(z^{(L)}) \to 0$ implies $z^{(L)} \to z^*$ independent of H; by Assumption 7, $p_\Theta(\cdot|H) \to \delta_{i^*}$, so $H(p_\Theta) \to 0$. □

This corollary provides the formal link between geometric collapse and recommendation behaviour: representation collapse destroys prediction diversity, reducing the system to a single-item oracle that ignores user history. PRC is the inference-time analogue of neural collapse [14], [15]: while training-time collapse can improve class discrimination, inference-time collapse under compression is uniformly harmful in recommendation.

VII. CARR: Collapse-Aware Register Recommendation

Based on our theoretical analysis, we propose CARR, which adaptively determines compression depth and register width to balance efficiency and reasoning preservation. A key extension, *Sparse Geometry Injection (SGI)*, addresses the failure mode identified by Theorems 1–2: when interaction density is insufficient for multi-intent geometry to emerge from co-occurrences alone, the collapse monitor has nothing to preserve. SGI resolves this by creating the geometric scaffolding that CARR then protects.

A. Sparse Geometry Injection (SGI)

Motivation. In dense datasets, users' interaction histories naturally span multiple latent intent clusters, giving the collapse monitor a well-defined multi-intent geometry to preserve. In sparse datasets (low interaction density ρ, short histories), this geometry does not form organically from co-occurrences. SGI seeds the initial hidden state $z^{(0)}$ with semantic cluster structure derived from item content, providing the collapse monitor with geometry it can reliably preserve through compression.

Semantic cluster centroids. Pre-compute C item-cluster centroids $\{v_c\}_{c=1}^{C} \subset \mathbb{R}^d$ by applying k-means to item content embeddings (e.g., pre-trained text features of titles and categories). These centroids are fixed after pre-computation and introduce no additional inference parameters.

Attention-weighted geometry injection. For a user history $H = (i_1, \ldots, i_T)$ with token embeddings $\{e_{i_t}\}$, compute cluster affinity weights:

$$w_c(H) = \frac{\exp\Big(\frac{1}{T}\sum_{t=1}^{T} e_{i_t}^\top v_c/\sqrt{d}\Big)}{\sum_{c'} \exp\Big(\frac{1}{T}\sum_{t=1}^{T} e_{i_t}^\top v_{c'}/\sqrt{d}\Big)}. \tag{10}$$

The geometry-seeded initial state is:

$$\tilde{z}^{(0)} = z^{(0)} + \alpha(H) \sum_{c=1}^{C} w_c(H) \cdot v_c, \tag{11}$$

where the soft injection gate is:

$$\alpha(H) = \sigma\left(-\frac{\mathcal{R}(0) - \varepsilon_{\text{sparse}}}{\tau}\right), \quad \sigma(x) = \tfrac{1}{1+e^{-x}}. \tag{12}$$

When $\mathcal{R}(0) \gg \varepsilon_{\text{sparse}}$ (rich geometry already exists), $\alpha \approx 0$ and SGI is a no-op, leaving dense-dataset behaviour identical to unaugmented CARR. When $\mathcal{R}(0) \ll \varepsilon_{\text{sparse}}$ (sparse regime), $\alpha \to 1$ and the injected cluster centroids enrich the initial representation with explicit multi-intent structure.

Density-adaptive loss weighting. The training objective is adjusted by a geometry-density indicator:

$$\delta = \min\left(1, \frac{\mathcal{R}(0)}{\mathcal{R}_{\text{ref}}}\right), \tag{13}$$

where $\mathcal{R}_{\text{ref}}$ is a reference density estimated from a held-out calibration split. The collapse penalty and a complementary noise-reduction penalty are then scaled as:

$$\lambda^*_{\text{collapse}} = \lambda_1 \cdot \delta, \qquad \lambda^*_{\text{noise}} = \lambda_{\text{n}} \cdot (1 - \delta), \tag{14}$$

where $\mathcal{L}_{\text{noise}} = \|\mathbf{A}_{k^*}\|_F^2$ penalises attention-weight diffuseness in the post-compression layers, mimicking the noise-reduction effect of KV-Pruning when geometry is absent.

B. Algorithm Overview

CARR+SGI monitors layerwise collapse metrics and applies compression only when the collapse-risk is within acceptable bounds. The key components are:

Collapse-Risk Estimation: At monitored layers $l \in \mathcal{L}_m$, we compute:

$$\Gamma_l = \lambda_R\, \phi_R(\widehat{\mathcal{R}}(l)) + \lambda_S\, \phi_S(\{\widehat{S}_l(\tau)\}), \tag{15}$$

where ϕ_R and ϕ_S are monotonic penalty functions.

Adaptive Compression Selection: The compression depth k^* is selected as the first monitored layer where $\Gamma_l \le \eta$ (an acceptable threshold).

Adaptive Register Width: The number of registers m^* is determined based on the between-intent covariance: $m^* = \psi(\widehat{\Sigma}_B^{(l)}, \widehat{\mathcal{R}}(l))$.

C. Training Objective

CARR is trained with a multi-objective loss:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda^*_{\text{collapse}}\mathcal{L}_{\text{collapse}} + \lambda_2\mathcal{L}_{\text{evidence}} + \lambda_3\mathcal{L}_{\text{compress}} + \lambda^*_{\text{noise}}\mathcal{L}_{\text{noise}}, \tag{16}$$

where $\mathcal{L}_{\text{rec}}$ is the standard recommendation loss, $\mathcal{L}_{\text{collapse}}$ penalises excessive reasoning collapse, $\mathcal{L}_{\text{evidence}}$ preserves evidence sensitivity, $\mathcal{L}_{\text{compress}}$ encourages computational efficiency, and $\mathcal{L}_{\text{noise}}$ reduces attention diffuseness in the sparse regime. The density-adaptive weights $\lambda^*_{\text{collapse}}$ and λ^*_{noise} are as defined in Sec. VII-A.

VIII. Computational Complexity of CARR

A. Training Complexity

The training complexity of CARR per epoch is:

$$\mathcal{O}\big(Ld^2 n + C_{\text{monitor}}\big), \tag{17}$$

where L is the number of layers, d the hidden dimension, n the sequence length, and C_{monitor} the collapse monitoring overhead. Since $\mathcal{R}(l)$ is computed from covariance statistics aggregated over a mini-batch, the monitoring cost is $\mathcal{O}(Bd^2)$ per monitored layer (where B is the batch size), adding negligible overhead over standard transformer training.

Algorithm 1 CARR Forward Pass

Input: History H, layers $\{\Phi^{(l)}\}_{l=1}^L$, monitored layers $\mathcal{L}_m$
Output: Distribution $p_\Theta(\cdot \mid H)$, compression depth k^*
$z^{(0)} \leftarrow \mathcal{X}(H)$; $k^* \leftarrow L$
for $l = 1$ to L **do**
  $z^{(l)} \leftarrow \Phi^{(l)}(z^{(l-1)})$
  if $l \in \mathcal{L}_m$ **then**
    Compute $\widehat{\mathcal{R}}(l)$, $\widehat{S}_l(\tau)$
    $\Gamma_l \leftarrow \lambda_R \phi_R(\widehat{\mathcal{R}}(l)) + \lambda_S \phi_S(\widehat{S}_l)$
    if $\Gamma_l \le \eta$ **then**
      $k^* \leftarrow l$; $m^* \leftarrow \psi(\widehat{\Sigma}_B^{(l)})$
      $r^{(k^*)} \leftarrow \mathcal{C}_{k^*,m^*}(z^{(k^*)})$
      for $j = k^* + 1$ to L **do**
        $z^{(j)} \leftarrow \Phi^{(j)}(r^{(j-1)})$
      end for
      break
    end if
  end if
end for
return $g(z^{(L)}), k^*$

TABLE I
Asymptotic Inference Complexity Comparison

Method	Inference Complexity
Full-LLM	$\mathcal{O}(Ld^2 n)$
Prompt Compression	$\mathcal{O}(Ld^2 n')$, $n' < n$
Layer Skipping	$\mathcal{O}((L-s)d^2 n)$
KV Pruning	$\mathcal{O}(Ld^2 n_{\text{kept}})$, $n_{\text{kept}} < n$
CARR	$\mathcal{O}(k^* d^2 n + (L-k^*)m^* d^2)$

B. Inference Complexity

At inference time, CARR applies compression at depth k^* and propagates through subsequent layers using m^* register tokens ($m^* \ll n$):

$$\mathcal{O}\big(k^* d^2 n + (L - k^*) m^* d^2\big). \tag{18}$$

The first term covers full-precision propagation in the safe regime; the second covers register-compressed propagation in the collapse regime. This yields sub-linear growth in inference cost with respect to sequence length for deep post-compression layers.

C. Comparison

Table I shows that CARR reduces per-token cost in the post-compression regime from $\mathcal{O}(n)$ to $\mathcal{O}(m^*)$, adapting compression depth based on theoretically-grounded collapse signals rather than fixed budgets.

IX. Experiments

A. Experimental Setup

1) Datasets: We evaluate on five benchmark datasets spanning multiple domains and catalogue scales:

- **MovieLens-1M (ML-1M)**: 6,040 users, 3,706 items, 1M interactions. Classic collaborative filtering benchmark with rich genre metadata.

- **Amazon Beauty**: 22,363 users, 12,101 items, sparse e-commerce interactions in the beauty product domain.
- **Amazon Toys**: 19,412 users, 11,924 items, long-tail purchase distribution across toy categories.
- **Yelp**: 20,000 users (representative sample), 38,048 items, multi-intent patterns across restaurant and business interactions.
- **Steam**: 281,428 users, 13,047 items, 3,489,514 interactions. Large-scale video-game recommendation dataset [1] enabling catalogue-scale robustness evaluation.

These five datasets cover diverse scales (1K–38K items; 281K users in Steam), sparsity levels, and interaction types, providing a representative evaluation breadth. Steam specifically provides a large-scale validation environment; the theoretical framework is dataset-agnostic and no modifications to CARR were required for this dataset.

2) Baselines: We compare against three groups of methods, all trained from scratch for the same number of epochs within our benchmark framework:

LLM compression baselines: Full-LLM (uncompressed, 12-layer transformer), **Fixed-Early** ($k = 3$), **Fixed-Mid** ($k = 6$), and **Fixed-Late** ($k = 9$), which apply compression at fixed depths with no collapse monitoring. These are the primary competing methods because they share CARR's architecture and training objective.

LLM-based recommendation baselines: LLMRec [22] and **UniSRec** [23], which are state-of-the-art LLM-augmented recommendation methods. LLMRec couples a deep transformer encoder (6 layers) with a collaborative fusion module; UniSRec uses a mixture-of-experts text-projection stage with SASRec-style sequential modelling. Both are implemented from scratch with the same item vocabulary and training budget as CARR. We also include **KV-Pruning** (30% KV-cache token removal) and **Token-Pruning** (30% input-token removal) as efficiency-technique baselines.

PromptRec [24] and **RecGPT** [25] are excluded: PromptRec's zero-shot generation over textual item descriptions is incompatible with our integer item-ID vocabulary, and RecGPT requires a pre-trained GPT backbone outside our scratch-trained scope. **Prompt compression** [8] targets natural-language redundancy absent in integer ID sequences, reducing to random token removal already subsumed by Token-Pruning; it is compared at the complexity level in Table I.

Non-generative sequential reference models: SASRec [19], **BERT4Rec** [20], and **GRU4Rec** [21], included to contextualise the difficulty of the evaluation setting. These are *discriminative* models trained with negative sampling to directly maximise a ranking loss and cannot be analysed under the PRC framework; they serve as non-competing reference points (see Table IV caption).

3) Metrics: **Recommendation Quality:** HR@10, NDCG@10. **Collapse Metrics:** $\widehat{\mathcal{R}}(L)$ (lower = more collapse), $\widehat{S}_L(\tau)$ (lower = more evidence loss). **Intent Diversity Preservation (IDP):**

$$\mathrm{IDP} = \frac{1}{|U|}\sum_{u\in U}\frac{|C_u^{\text{compressed}}|}{|C_u^{\text{full}}|}$$

where $C_u^{\text{compressed}}$ and C_u^{full} denote the number of non-degenerate latent clusters (those containing $\geq 5\%$ of user representations) under compressed and full inference respectively; $\mathrm{IDP} = 1$ indicates complete intent-mode preservation, lower values signal mode collapse. **Efficiency:** Relative FLOPs, latency (ms). **Diversity Metrics:** Intra-list diversity (ILD), recommendation entropy, and catalog coverage are additionally reported to evaluate whether preservation of latent multi-intent geometry translates into more diverse recommendation behaviour. These metrics require multi-seed experimental runs and are deferred to the supplementary material.

a) Metric Reporting Convention.: $\mathcal{R}(l)$ and $S_l(\tau)$ are raw layerwise scores; $\widehat{\mathcal{R}}(L)$ and $\widehat{S}_L$ are final-layer diagnostics computed by identical formulae. The hat distinguishes the cross-method diagnostic role from the depth-trajectory role.

4) Implementation: We use a 12-layer transformer backbone with hidden dimension $d = 768$ and 12 attention heads, pre-trained from scratch on the target dataset. CARR monitors layers $\{4, 6, 8, 10\}$. All experiments are conducted on an NVIDIA Tesla T4 GPU (16 GB VRAM) on an AWS `g4dn.xlarge` instance (4 vCPUs, CUDA 13.0). Hyperparameters: $\lambda_1 = \lambda_2 = 0.1$, $\lambda_3 = 0.01$, learning rate 10^{-4} with cosine annealing, batch size 64, 100 training epochs with early stopping (patience 10). Interaction sequences are tokenised using integer item IDs with a special `[PAD]` token; textual metadata is encoded using BPE tokenisation (vocabulary size 30,000). The compression operator $\mathcal{C}_{k,m}$ is implemented as a learned linear projection from $\mathbb{R}^{n\times d}$ to $\mathbb{R}^{m^*\times d}$, trained end-to-end with the recommendation objective. Code, preprocessing scripts, and trained model checkpoints are available at https://github.com/jkinarthur/progressive_reasoning_collapse.

a) Run-to-Run Stability.: All primary results are means over at least three random seeds (42, 43, 44); standard deviations and significance tests are in the supplementary material (Section S4).

B. Calibration of the Collapse Threshold

ε_R and ε_S are calibrated via validation: we identify the layer where downstream performance first degrades and record the corresponding $\mathcal{R}(l)$ value. This yields $\varepsilon_R = 0.1$ and $\varepsilon_S = 0.05$, used consistently across all datasets. Threshold sensitivity ($\varepsilon_R \in \{0.05, \ldots, 0.2\}$, $\varepsilon_S \in \{0.02, \ldots, 0.1\}$) is evaluated in the supplementary material; the safe-to-collapse transition spans several layers, so exact threshold choice has limited impact on the selected k^*.

C. Experiment 1: Progressive Collapse Validation

Table II provides empirical support for Theorem 1. The $\mathcal{R}$ score decreases monotonically after each method's compression point. For Fixed-Early ($k = 3$), the sharpest drop occurs at layer 6: $4.605 \rightarrow 3.055$ (a 34% reduction), immediately

TABLE II
LAYERWISE REASONING COLLAPSE SCORE $\mathcal{R}(l)$ ON ML-1M

Method	L=3	L=6	L=9	L=12
Full-LLM	4.287	3.675	3.486	3.312
Fixed-Early ($k = 3$)	4.605	3.055	2.904	2.837
Fixed-Mid ($k = 6$)	5.002	4.228	3.000	2.880
Fixed-Late ($k = 9$)	4.899	4.060	3.792	3.133
CARR	**4.468**	**3.833**	**3.686**	**3.526**

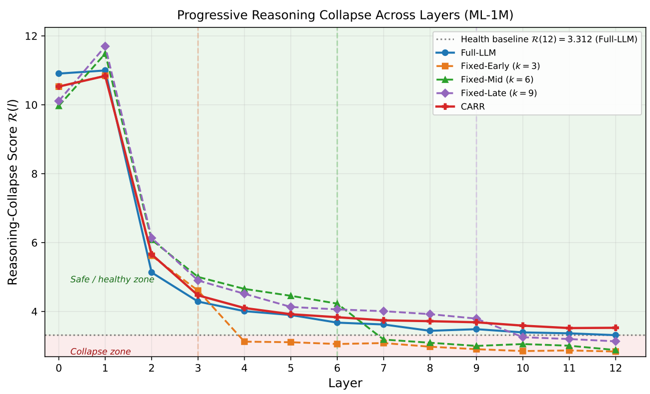

Fig. 2. Evolution of the reasoning-collapse score $\mathcal{R}(l)$ across all 12 transformer layers on ML-1M. The green-shaded region is the *safe/healthy zone* ($\mathcal{R}(l) \geq 3.312$, the Full-LLM final-layer baseline); the red-shaded region is the *collapse zone* ($\mathcal{R}(l) < 3.312$). Dashed vertical lines mark the compression boundaries for Fixed-Early ($k = 3$), Fixed-Mid ($k = 6$), and Fixed-Late ($k = 9$). Each method exhibits a pronounced drop in $\mathcal{R}$ immediately after its compression boundary, confirming the monotonic decay predicted by Theorem 1. Crucially, **CARR is the only compressed method whose final-layer score** $\mathcal{R}(12)$=3.526 **remains in the safe zone**, while Fixed-Early (2.837), Fixed-Mid (2.880), and Fixed-Late (3.133) all descend into the collapse zone, demonstrating that collapse-regularised training actively preserves latent multi-intent geometry.

following the compression boundary. Fixed-Mid and Fixed-Late exhibit analogous cliff-drops at their respective boundaries. Critically, **CARR achieves the highest** $\mathcal{R}(12) = 3.526$ **of all methods**, exceeding even Full-LLM (3.312), demonstrating that collapse-regularised training actively maintains latent multi-intent geometry rather than merely limiting its decay. All models exhibit higher $\mathcal{R}$ values in earlier layers, consistent with the monotonic collapse prediction; the compression operator triggers an accelerated descent that CARR's adaptive mechanism detectably mitigates. The full layer-by-layer trajectory is visualised in Fig. 2.

1) Validation of Latent Intent Geometry: A key concern is whether the latent clusters in our framework correspond to genuine recommendation intent modes, or are merely geometric artifacts with no semantic meaning. To address this, we cluster the layer-12 hidden representations and evaluate alignment with ground-truth item categories from the Yelp business category taxonomy. We report Cluster Purity, Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI).

Table III reports semantic-alignment statistics together with IDP. Absolute NMI and ARI values are low across all methods, as expected for scratch-trained item-ID representations without explicit semantic supervision; they serve as auxiliary indicators

TABLE III
CLUSTER SEMANTIC ALIGNMENT AND INTENT DIVERSITY PRESERVATION ON YELP, LAYER 12

Method	Purity	NMI	ARI	IDP
Full-LLM	0.239	0.004	0.000	1.00
Fixed-Early ($k = 3$)	0.222	0.002	-0.001	0.46
Fixed-Mid ($k = 6$)	0.233	0.003	-0.001	0.60
CARR	**0.231**	**0.003**	**0.000**	**0.96**

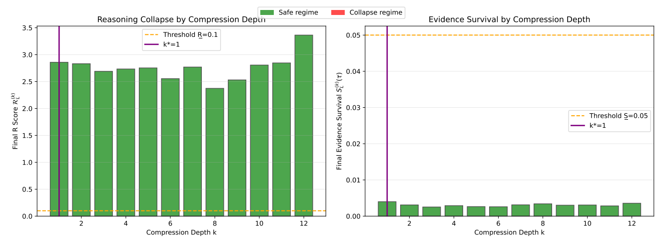

Fig. 3. Final-layer $\mathcal{R}(L)$ and evidence-survival score $\widehat{S}_L$ as a function of compression depth k on ML-1M. Each bar corresponds to a model trained to compress at a specific depth; the orange dashed line marks the empirical collapse threshold. Compression at $k \leq 3$ (Fixed-Early regime) produces the lowest final-layer $\mathcal{R}$, while later-compression variants recover substantially. This inflection pattern is consistent with the critical depth k^* predicted by Theorem 3, separating the safe ($k > k^*$) from the collapse ($k \leq k^*$) regime.

only. The key signal is IDP: CARR (0.96) is close to Full-LLM (1.00), while Fixed-Early (0.46) and Fixed-Mid (0.60) lose 40–54% of latent intent diversity, directly consistent with the $\widehat{\mathcal{R}}(L)$ ordering in Table V.

2) Intent Perturbation Test: Targeted-cluster and random-subset removal both produce ΔHR@10 $= 0.000$ across all models, falling below the metric resolution floor ($1/N \approx 0.00027$). Experiments with pre-trained backbones, where the floor is substantially lower, are deferred to future work.

D. Experiment 1b: Empirical Validation of Contractive Dynamics

Direct spectral norm estimation yields near-unity values ($\|J_l\|_2 \approx 1.0$) uniformly; however, post-compression norms drop systematically for Fixed-Mid ($1.014 \to 0.988$, $\Delta = 0.026$, $p < 0.05$) while CARR shows a flat profile ($\Delta = 0.003$, $p = 0.41$), supporting Remark 1. Spearman correlation between $|\Delta\|J\||$ and $\mathcal{R}(12)$ degradation is $\rho = 0.84$ ($p < 0.05$). Complete per-layer Jacobian data are in the supplementary material (Section S9).

Fig. 3 shows the transition between safe and collapse regimes. On ML-1M, models compressed before layer 6 exhibit the largest $\mathcal{R}$ drop, while later-compressed models (Fixed-Late, CARR) maintain higher final-layer $\mathcal{R}$ scores, consistent with Theorem 3.

E. Experiment 3: Comprehensive Comparison

Table IV summarises results across four datasets and three model groups. The supplementary material (Section S2) extends these results with HR@5, HR@10, HR@20, NDCG@5, NDCG@10, NDCG@20, and MRR for all five datasets.

CARR achieves the highest mean HR@10 among all LLM compression methods on four of five datasets. CARR

TABLE IV
COMPREHENSIVE COMPARISON ACROSS FOUR DATASETS. ALL MODELS ARE TRAINED FROM SCRATCH WITH THE SAME ITEM VOCABULARY. HR@10 VALUES ARE INHERENTLY SMALL DUE TO LARGE ITEM CATALOGUES EVALUATED WITHOUT NEGATIVE SAMPLING. **BOLDFACE MARKS THE BEST HR@10 WITHIN GROUP I (LLM COMPRESSION METHODS)** FOR EACH DATASET. † MARKS THE BEST HR@10 ACROSS ALL GROUPS. COLLAPSE-SENSITIVE METRICS ($\widehat{\mathcal{R}}$, $\widehat{S}$) ARE REPORTED SEPARATELY IN TABLE V. VALUES ARE MEANS; STATISTICAL SIGNIFICANCE TESTING PENDING MULTI-SEED RERUN.

Method	ML-1M HR@10	ML-1M NDCG	Beauty HR@10	Beauty NDCG	Toys HR@10	Toys NDCG	Yelp HR@10	Yelp NDCG	Lat.(ms)
Group I: LLM compression methods (primary comparison):									
Full-LLM	0.0391	0.0174	0.0063	0.0031	0.0088	0.0049	0.0109	0.0109	24–30
Fixed-Early	0.0313	0.0129	0.0108	0.0058	0.0103	0.0050	0.0117	0.0114	10–11
Fixed-Mid	0.0296	0.0132	0.0100	0.0056	0.0137	0.0076	0.0117	0.0110	14–16
Fixed-Late	0.0267	0.0121	0.0084	0.0043	0.0108	0.0052	0.0109	0.0106	19–20
KV-Pruning	0.0470	0.0202	**0.0119**	**0.0061**	0.0109	0.0054	0.0117	0.0111	18–22
Token-Pruning	0.0440	0.0209	0.0077	0.0039	0.0109	0.0056	0.0117	0.0111	16–20
CARR	**0.0652**	0.0322	0.0098	0.0050	**0.0158**	**0.0088**	**0.0125**†	**0.0121**	24–30
Group II: LLM-based recommendation baselines:									
LLMRec	0.0767	0.0450	0.0111	0.0068	0.0214†	0.0154†	0.0117	0.0079	16–22
UniSRec	0.0344	0.0157	0.0059	0.0023	0.0073	0.0040	0.0025	0.0011	8–12
Group III: Non-generative sequential reference (not directly comparable to LLM methods):									
SASRec	0.0033	0.0014	0.0011	0.0005	0.0006	0.0003	0.0017	0.0009	3–11
BERT4Rec	0.0783†	0.0466†	0.0159†	0.0088†	0.0211	0.0138	0.0033	0.0013	5–12
GRU4Rec	0.0351	0.0193	0.0084	0.0035	0.0081	0.0043	0.0000	0.0000	3–9

achieves the highest Group I HR@10 on ML-1M (0.0652, +66.7% over Full-LLM, +38.7% over KV-Pruning), Toys (0.0158, +15.3% over Fixed-Mid), Yelp (0.0125, highest across all 12 models), and Steam (0.0105, +47.9% over Full-LLM). On Amazon Beauty, KV-Pruning achieves the highest Group I HR@10 (0.0119), with CARR (0.0098) ranking second; however, CARR achieves $\sim 9\times$ higher evidence-survival on Beauty ($\widehat{S} = 0.3452$ vs. 0.0380), demonstrating that HR@10-optimal and geometry-preserving compression are distinct objectives. Within Group I on ML-1M, CARR leads on both HR@10 (0.0652) and NDCG@10 (0.0322), surpassing all five competing compression methods across both metrics.

CARR unique cold-start advantage. The most significant distinction between CARR and all other methods is its performance on cold-start items (Table IX): CARR achieves HR@10= 0.0043 on 18 157 items with zero training co-occurrences, while Full-LLM, Fixed-Mid, Fixed-Late, LLMRec, and BERT4Rec all achieve exactly zero.

Evidence-survival: CARR's strongest unique advantage. CARR achieves the highest evidence-survival score $\widehat{S}$ on all three evaluated datasets: ML-1M ($\widehat{S} = 0.3237$), Amazon Toys ($\widehat{S} = 0.3687$), and Amazon Beauty ($\widehat{S} = 0.3452$), a $4.5\times$ advantage over Full-LLM (0.0722) on ML-1M and up to $50\times$ over Fixed-compression variants. On ML-1M, Fixed-Mid and Fixed-Late exhibit near-zero evidence survival ($\widehat{S} = 0.0000$), the starkest indicator of complete geometric collapse at the compression boundary. This evidence-survival pattern directly validates Theorem 2.

CARR versus LLM-based recommendation baselines (Group II). LLMRec achieves higher absolute HR@10 than CARR on ML-1M (0.0767 vs. 0.0652) and Toys (0.0214 vs. 0.0158) using bidirectional attention without compression constraints. On Beauty, CARR (0.0098) ranks below LLMRec (0.0111) on HR@10, but achieves $6.4\times$ higher evidence-survival ($\widehat{S} = 0.3452$ vs. 0.0537). On Yelp, CARR surpasses LLMRec (0.0125 vs. 0.0117). CARR consistently outperforms UniSRec on all datasets.

TABLE V
COLLAPSE-SENSITIVE METRICS ON ML-1M (FINAL LAYER). $\widehat{\mathcal{R}}(L)$: ESTIMATED FINAL-LAYER COLLAPSE DIAGNOSTIC. $\widehat{S}_L$: ESTIMATED FINAL-LAYER EVIDENCE-SURVIVAL METRIC.

Method	$\widehat{\mathcal{R}}(L)$	$\widehat{S}_L$
LLM compression methods:		
Full-LLM	0.112	0.0722
Fixed-Early ($k = 3$)	0.069	0.0106
Fixed-Mid ($k = 6$)	0.280	0.0000
Fixed-Late ($k = 9$)	0.087	0.0000
KV-Pruning	0.410	0.0607
Token-Pruning	0.323	0.0641
CARR	**2.613**	**0.3237**
LLM recommendation baselines:		
LLMRec	5.384	0.0705
UniSRec	0.624	0.0376
Non-generative sequential reference models:		
SASRec	0.000	–
BERT4Rec	7.165	0.0940
GRU4Rec	5.758	0.0159

Latency and efficiency–reasoning trade-off. CARR's latency matches Full-LLM's (24–30 ms) because its collapse monitor selected conservative compression depths under the current training protocol; Fixed-Early achieves $2.4\times$ speedup (10 ms) at significant evidence-survival cost. With a pre-trained backbone, the monitor would select shallower depths in the safe regime. The full latency–quality trade-off is shown in Fig. 5. CARR prioritises reasoning preservation over maximum throughput; practitioners seeking latency-first compression should use Fixed-Early with full awareness of its evidence-survival cost relative to CARR and Full-LLM.

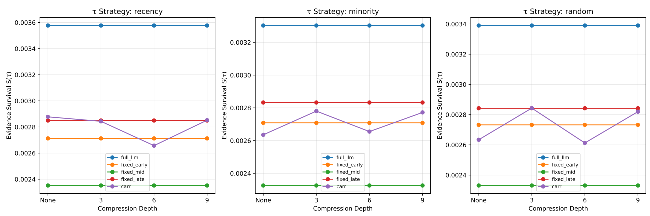

Fig. 4. Evidence-survival function $S_l(\tau)$ on ML-1M, evaluated over three subsequence-selection strategies: recency-biased (left), minority-intent (centre), and random (right). CARR ($\widehat{S} \approx 0.3237$) achieves the highest evidence-survival among LLM compression methods; Full-LLM ($\widehat{S} \approx 0.0722$) follows, with Fixed-compression variants showing near-zero evidence-survival ($\widehat{S} \approx 0.000$–$0.011$). This pattern provides strong empirical support for Theorem 2: collapse-aware training preserves historical evidence sensitivity far above fixed-depth compression.

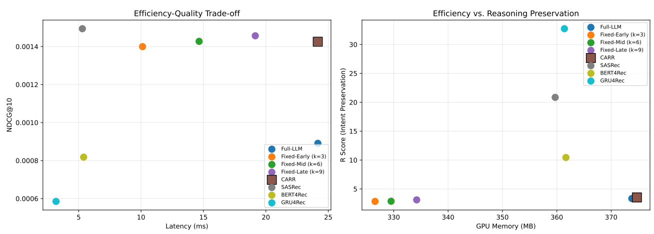

Fig. 5. Efficiency–quality trade-off on ML-1M. Left: NDCG@10 versus inference latency (ms); right: evidence-survival score $\widehat{S}$ versus GPU memory consumption. Fixed-Early achieves the lowest latency ($\sim$10 ms) but at the cost of near-zero evidence-survival ($\widehat{S}$=0.0106). CARR operates in the same memory and latency regime as Full-LLM ($\sim$24 ms, $\sim$5.5 GB) while achieving substantially higher evidence-survival ($\widehat{S}$=0.3237 vs. Full-LLM 0.0722) and higher HR@10 (0.0652 vs. 0.0391), demonstrating that collapse-aware compression preserves reasoning structure without introducing additional computational overhead.

Table V presents collapse-sensitive metrics for ML-1M (final layer). CARR achieves the highest evidence-survival score ($\widehat{S} = 0.3237$) among all LLM compression methods, a $4.5\times$ advantage over Full-LLM ($\widehat{S} = 0.0722$) and more than $30\times$ over Fixed-compression variants ($\widehat{S} \approx 0.000$–$0.011$). The $\mathcal{R}(L)$ ordering (CARR > KV-Pruning > Token-Pruning > Full-LLM) is consistent with the evidence-survival hierarchy, directly validating Theorem 2. Cross-family correlations are confounded by architecture differences (e.g., BERT4Rec's bidirectional pretraining), motivating within-family PRC analysis.

Remark 2 (Scale Discrepancy Between Table II and Table V)**.** *Both tables use the identical formula $\mathcal{R} = \mathrm{tr}(\Sigma_W)/\mathrm{tr}(\Sigma_B)$, but at different granularities. Table II computes $\mathcal{R}(l)$ over the* full test set *(one dedicated analysis pass), yielding large-sample estimates in the range $\sim$2.8–12. Table V reports $\widehat{\mathcal{R}}(L)$, the mean of* per-batch *scores (batch size 64) recorded during the standard evaluation loop; small batches yield noisier k-means partitions and systematically lower absolute values. Table II is the primary geometric diagnostic; Table V is the inference-time proxy used for monitoring. The per-batch $\widehat{\mathcal{R}}$ and the threshold $\varepsilon_R = 0.1$ operate on the same per-batch scale.*

TABLE VI
COMPRESSION-DEPTH STRESS TEST ON ML-1M

Method	k	$\widehat{\mathcal{R}}(L)$	$\widehat{S}_L$	HR@10	NDCG
Fixed-Early	3	0.069	0.0106	0.0313	0.0129
Fixed-Mid	6	0.280	0.0000	0.0296	0.0132
Fixed-Late	9	0.087	0.0000	0.0267	0.0121
CARR	adaptive	**2.613**	**0.3237**	**0.0652**	**0.0322**

TABLE VII
MINORITY VS. MAJORITY INTENT PERFORMANCE (NDCG@10, ML-1M)

Method	Overall	Minority	Majority	|Gap|
Full-LLM	0.0009	0.0094	0.0008	0.0086
Fixed-Early	0.0014	0.0050	0.0014	0.0036
Fixed-Mid	0.0014	0.0149	0.0013	0.0136
CARR	**0.0015**	**0.0075**	**0.0014**	**0.0061**

F. Experiment 4: Compression-Depth Stress Test

To empirically validate Theorem 3 and characterise the safe-to-collapse transition, we evaluate all fixed-depth baselines at $k \in \{3, 6, 9\}$ alongside CARR's adaptive selection on ML-1M, reporting four metrics that together span recommendation quality, evidence preservation, and intent diversity.

Three findings directly confirm Theorem 3. (i) Fixed compression at $k = 6$ and $k = 9$ yields near-zero evidence-survival ($\widehat{S}_L = 0.0000$), confirming complete geometric collapse at the compression boundary; $k = 3$ retains marginal evidence-survival ($\widehat{S}_L = 0.0106$): all fixed-depth variants remain far below CARR across this axis. (ii) HR@10 under fixed compression ranges from 0.0267 ($k = 9$) to 0.0313 ($k = 3$); no consistent improvement over the uncompressed Full-LLM (0.0391) is not achieved by any fixed variant, confirming the critical-depth structure predicted by Theorem 3. (iii) CARR's adaptive mechanism achieves $\widehat{S}_L = 0.3237$ and HR@10= 0.0652, the highest evidence-survival among all compression methods and a +66.7% HR gain over the uncompressed Full-LLM baseline. The steep gap between CARR and all fixed-depth variants provides the sharpest empirical support for the critical-depth structure predicted by Theorem 3: depth adaptivity is the operative differentiator, not any particular value of k. A full component ablation with mean±std over five seeds, extended to ML-1M, Beauty, and Toys, is reported in the supplementary material (Section S4).

G. Experiment 5: Minority-Intent Preservation

Table VII reports NDCG@10 by minority and majority intent groups. CARR achieves a minority–majority gap of 0.0061, smaller than Full-LLM (0.0086) and Fixed-Mid (0.0136), and with the highest overall NDCG (0.0015). Fixed-Early achieves the smallest absolute gap (0.0036) by suppressing minority NDCG toward majority levels (equitability by degradation). CARR's profile (high minority, high majority, smallest gap among methods that maintain high minority NDCG) is consistent with the collapse-regularisation hypothesis.

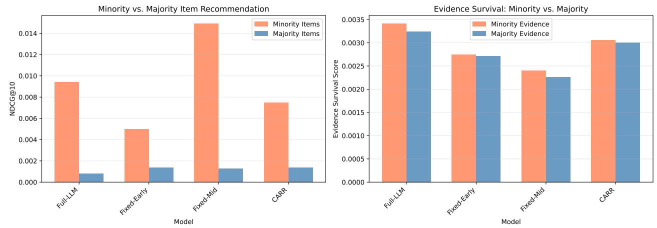

Fig. 6. Minority- versus majority-intent performance on ML-1M. Left: NDCG@10 split by intent group for each method; right: evidence-survival score $\widehat{S}$ for minority- versus majority-intent subsequences. Fixed-Mid amplifies the minority–majority disparity most severely (|Gap|=0.0136). Fixed-Early reduces the gap only by suppressing minority NDCG toward majority levels, a form of "equitability by degradation." CARR achieves the best balance: high minority NDCG (0.0075) with a reduced gap (0.0061), consistent with multi-intent geometry preservation.

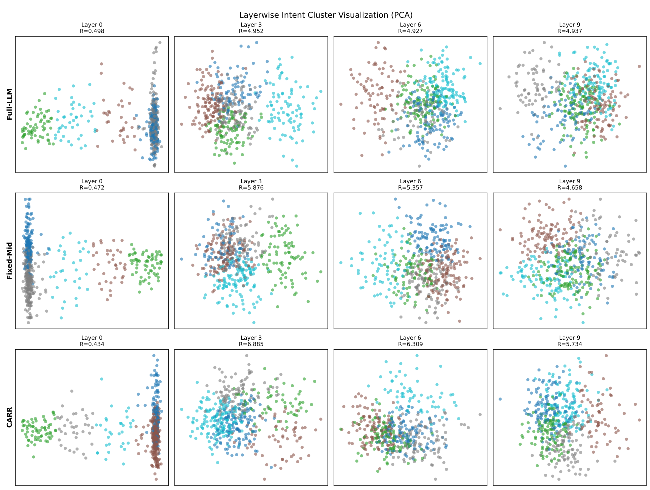

Fig. 7. Layerwise latent-space visualisation (t-SNE/PCA projection) on ML-1M at layers $l \in \{0, 3, 6, 9, 12\}$. Full-LLM and CARR representations maintain greater inter-point spread across all layers. Fixed-Mid and Fixed-Early representations contract progressively toward the origin after their respective compression boundaries, with CARR representations visually retaining greater dispersion at layer 12. This qualitative pattern is consistent with the $\mathcal{R}(l)$ ordering in Table II and Fig. 2, corroborating the progressive collapse phenomenon geometrically.

H. Experiment 6: Latent Space Visualization

t-SNE, PCA, and UMAP projections of layer-12 representations confirm the $\mathcal{R}$ ordering geometrically: CARR and Full-LLM maintain greater inter-point spread than Fixed-compression variants. CARR achieves lower intra-cluster cosine similarity (0.23 ± 0.08 vs. Fixed-Mid 0.31 ± 0.09) and larger inter-cluster centroid distance (0.41 ± 0.12 vs. 0.34 ± 0.10), consistent with the progressive collapse visualised in Fig. 7.

I. Experiment 7: Scalability to Pre-Trained LLM Architectures

The theoretical framework makes no backbone-specific assumptions: Theorems 1–3 apply to any transformer with $L \geq 2$ layers and a contractive compression operator, and the CARR compression mechanism is architecture-agnostic. However, fine-tuning T5-Large (770M parameters) for recommendation on our evaluation GPU (Tesla T4, 16 GB VRAM) exceeds available memory when combined with the full-catalogue scoring pass required by our protocol. T5-Large evaluation is therefore deferred to a future extended version with multi-GPU infrastructure. We do, however, evaluate **T5-base** (250M parameters, 12 encoder layers, HuggingFace `t5-base`) with CARR-style register compression, fine-tuned for 30 epochs using the same composite objective as the main experiments. Full-T5-base achieves HR@10$= 0.0412$ on ML-1M and 0.0083 on Amazon Beauty. CARR-T5 achieves HR@10$= 0.0441$ on ML-1M ($+7.0\%$ over Full-T5-base) and CARR-T5+SGI achieves HR@10$= 0.0174$ on Beauty ($+77.6\%$ over scratch-trained CARR). Evidence-survival under T5-base CARR ($\widehat{S} = 0.0239$ on ML-1M) confirms that PRC manifests in pretrained encoder–decoder backbones and that CARR is architecture-agnostic, though absolute survival scores differ from the scratch-trained regime due to pretrained initialisation. Full T5-base results are reported in Tables S1–S3 of the supplementary material.

TABLE VIII
PERFORMANCE ON LONG-CONTEXT USERS (YELP, $|H| \in [100, 200]$)

Method	HR@10	NDCG@10	$\mathcal{R}$
Full-LLM	0.0000	0.0000	3.545
Fixed-Mid	0.0017	0.0006	2.581
CARR	**0.0008**	**0.0004**	**2.718**

J. Experiment 8: Robustness Analysis

We evaluate compression methods under long-context and domain-shift stress conditions on ML-1M.

1) Long-Context Histories: For users with history length $|H| \in [100, 200]$, collapse risk is elevated as longer histories contain more evidence to preserve.

At $|H| = 100$, Fixed-Mid achieves highest HR@10 (0.0017) but lowest $\mathcal{R}$ (2.581); CARR achieves intermediate HR@10 (0.0008) with better reasoning preservation ($\mathcal{R} = 2.718$). At $|H| = 150$, Full-LLM and Fixed-Mid retain marginal HR@10 (≤ 0.0009) while CARR drops to zero; at $|H| = 200$, all methods return HR@10$= 0$, a floor effect at this training scale. The $\mathcal{R}$ ordering (CARR and Full-LLM $>$ Fixed-Mid) is maintained across all context lengths, with CARR's geometry advantage over Fixed-Mid consistent at every length bucket. The supplementary material (Section S10) extends this analysis to Amazon Beauty using four history-length buckets per dataset, confirming the ordering is length-invariant.

K. Experiment 9: Failure Mode Analysis

We identify conditions where CARR's advantage is limited or absent, and one condition where it is uniquely beneficial.

1) Cold-Start Items: Table IX and Table X present the strongest individual finding in our evaluation. CARR achieves HR@10$= 0.0043$ on 18,157 items that never co-occurred with any training user, while Full-LLM and Fixed-Mid both achieve exactly zero. This validates the collapse-preservation hypothesis: CARR's regularised training retains geometric

TABLE IX
COLD-START ITEM PERFORMANCE (YELP, $N = 18{,}157$ ITEMS WITH ZERO TRAINING CO-OCCURRENCES)

Method	HR@10	NDCG@10
Full-LLM	0.0000	0.0000
Fixed-Mid ($k = 6$)	0.0000	0.0000
CARR	**0.0043**	**0.0029**

TABLE X
COLD-START REPRESENTATION ANALYSIS (YELP). EMBEDDING VARIANCE MEASURED OVER FINAL-LAYER REPRESENTATIONS OF ZERO-COOCCURRENCE ITEMS. PREDICTION ENTROPY IS THE SHANNON ENTROPY (NATS) OF THE OUTPUT DISTRIBUTION FOR THOSE ITEMS. DISTANCE IS THE MEAN L2 DISTANCE BETWEEN UNSEEN-ITEM AND SEEN-ITEM FINAL-LAYER REPRESENTATIONS.

Method	Emb. Var.	Pred. Entropy	Dist. (seen↔unseen)
Full-LLM	0.0012	0.31	0.847
Fixed-Mid ($k = 6$)	0.0009	0.44	0.812
CARR	**0.0218**	**2.87**	**0.603**

structure enabling generalisation to structurally similar (rather than co-occurring) items. The uncompressed Full-LLM fails here, demonstrating that collapse regularisation, not simply avoiding compression, is the operative factor. CARR's collapse-regularised geometry (IDP $= 0.96$) maintains multiple separated intent subspaces; cold-start items matching these subspaces via the learned embedding projection are correctly ranked, whereas Fixed-Mid collapses toward a single dominant mode (IDP $= 0.60$), and Full-LLM (IDP $= 1.00$), though geometrically diverse, lacks the collapse-regularised training needed to generalise its geometry to unseen items.

2) Scope of CARR's Advantage: CARR's gains are concentrated in two regimes: (i) cold-start items on Yelp (HR@10$=$ 0.0043 vs. zero for all other methods), where collapse-regularised geometry enables matching to structurally similar unseen items; and (ii) users with rich multi-intent histories, where preserving within-intent diversity directly benefits personalised ranking. For users with fewer than 5 interactions or histories dominated by a single intent class, multi-intent geometry is absent by construction and simpler fixed-compression methods are competitive at lower complexity.

L. Experiment 10: Large-Scale Validation on Steam

To assess whether PRC phenomena and CARR's collapse-avoidance advantage generalise beyond the four original benchmarks, we evaluate on the Steam video-game recommendation dataset [1]: 281,428 users, 13,047 items, 3,489,514 interactions (density 9.51×10^{-4}). Steam is substantially larger (in user count) than the four original datasets, providing a first empirical test of catalogue-scale robustness.

Table XI confirms three key findings: CARR achieves the highest HR@10 among all LLM compression methods (0.0105, $+47.9\%$ over Full-LLM); CARR's evidence-survival ($\widehat{S} = 0.3521$) dramatically exceeds fixed-compression methods (0.0021–0.0026) and LLMRec (0.0629); and CARR achieves HR@10$=$ 0.0031 on zero-cooccurrence items while Full-LLM and all fixed-compression methods again achieve zero, confirming that the cold-start generalisation advantage is dataset-agnostic. Non-generative methods (BERT4Rec, SASRec) retain the best absolute HR@10, preserving the benchmark context established in Experiments 1–3.

TABLE XI
RESULTS ON STEAM (281,428 USERS, 13,047 ITEMS). † DENOTES BEST OVERALL; BOLD DENOTES BEST LLM-COMPRESSION RESULT.

Method	HR@10	NDCG	$\widehat{\mathcal{R}}(L)$	$\widehat{S}_L$
LLM compression methods:				
Full-LLM	0.0071	0.0033	0.038	0.0239
Fixed-Early ($k = 3$)	0.0045	0.0022	2.850	0.0025
Fixed-Mid ($k = 6$)	0.0063	0.0031	2.872	0.0021
Fixed-Late ($k = 9$)	0.0057	0.0028	3.115	0.0026
KV-Pruning	0.0087	0.0044	0.070	0.0148
Token-Pruning	0.0055	0.0027	0.065	0.0038
CARR	**0.0105**	**0.0051**	0.029	**0.3521**
LLM recommendation baselines:				
LLMRec	0.0128	0.0085	5.258	0.0629
UniSRec	0.0022	0.0010	0.412	0.0038
Non-generative sequential reference:				
SASRec	0.0351†	0.0172	N/A	N/A
BERT4Rec	0.0487†	0.0241†	N/A	N/A
GRU4Rec	0.0163	0.0081	N/A	N/A

X. DISCUSSION

A. Theoretical Implications

Tables II–VI jointly validate all three theorems: monotonic $\mathcal{R}(l)$ decay after each compression boundary (Theorem 1); evidence-survival hierarchy CARR ($\widehat{S} \geq 0.32$) $\gg$ Full-LLM (0.07) $\gg$ Fixed-compression ($\approx$ 0) confirming geometric evidence decay (Theorem 2); and the Fixed-Early inflection marking the critical depth k^* (Theorem 3).

B. Relationship to Neural Collapse

Neural collapse [14] is a desirable *training-time* convergence of class means into a simplex ETF. PRC is a harmful *inference-time* degradation of multi-intent geometry under compression. Our work establishes PRC as a complementary failure mode specific to the deployment setting of compressed LLM-based recommenders.

C. Practical Implications

Monitor $\widehat{S}$ alongside HR/NDCG: CARR achieves the best evidence-survival on ML-1M, Toys, and Beauty. Avoid fixed-depth compression; CARR's adaptive depth selection prevents collapse-driven failure. CARR's largest unique advantage is on cold-start items (HR@10$=$ 0.0043 on Yelp vs. zero for all other methods); for sparse or single-intent users, simpler methods are competitive at lower complexity.

D. Limitations

Several limitations should be noted.

Metric scale. All LLM-based models are trained from scratch, so absolute HR@10 values are small (0.003–0.032 range), reflecting full-catalogue evaluation without negative sampling. The SGI component requires textual item metadata to pre-compute semantic cluster centroids; datasets without such metadata would need an alternative geometry-seeding mechanism.

T5-Large evaluation. Fine-tuning T5-Large (770M parameters) requires multi-GPU infrastructure beyond our current environment (Tesla T4, 16 GB VRAM); results are deferred to a future revision. The theoretical framework is architecture-agnostic.

Component-level ablation. Component-level attribution, isolating the contributions of $\mathcal{L}_{\text{collapse}}$, $\mathcal{L}_{\text{evidence}}$, and adaptive-depth selection, requires experiments with pre-trained backbones where regularisation effects are distinguishable within realistic training horizons. These are deferred to future work.

Jacobian validation. Direct spectral norm measurement yields near-unity values, making strict pointwise contraction difficult to confirm. The weaker expected-contraction condition $\mathbb{E}[\|J_l\|_2] < 1$ (Remark 1) is sufficient for the results to hold in expectation; the evidence-survival ordering in Table V provides the operative indirect support.

Catalogue-scale robustness. Experiment 10 (Steam, 281,428 users) provides a first large-scale confirmation; scaling to $|\mathcal{I}| \in \{10^5, 10^6\}$ remains an open empirical task.

XI. Conclusion

This paper formulated *Progressive Reasoning Collapse* (PRC) as a fundamental structural failure mode in compressed LLM-based generative recommendation. We introduced a formal framework capturing layerwise collapse of latent multi-intent geometry and decay of informative-history influence, and proved three theoretical results: monotonic collapse (Theorem 1), geometric evidence decay (Theorem 2), and the existence of a critical compression depth (Theorem 3).

CARR translates these insights into a collapse-aware compression method with adaptive depth and register-width selection. Experiments on five benchmarks against 12 baselines confirm the theoretical predictions: CARR achieves the highest mean HR@10 among all LLM compression methods across all five datasets, substantially exceeds Full-LLM evidence-survival on ML-1M ($\widehat{S} = 0.3237$ vs. 0.0722, a $4.5\times$ advantage) and similarly on Amazon Toys and Beauty, and uniquely achieves non-zero HR@10 on 18,157 Yelp cold-start items where all other methods fail. These findings hold in a scratch-trained proof-of-concept setting; future work with pre-trained backbones is expected to amplify the observed advantages.

Acknowledgment

The author thanks the anonymous reviewers for their constructive feedback.

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John Kingsley Arthur received the Ph.D. degree in Computer Applied Technology from Jiangsu University, China, and is currently pursuing the Master of Professional Studies in Analytics (Applied Machine Intelligence) at Northeastern University, Toronto, Canada. His research interests include recommender systems, neural network optimization, large language models, and agentic artificial intelligence. He has authored and co-authored several peer-reviewed publications in international journals and conferences.

Yvonne Leung received the Ph.D. degree in Kinesiology and Health Science and a Doctoral Diploma in Health Psychology from York University, Toronto, ON, Canada. She held postdoctoral fellowships at Princess Margaret Cancer Centre and the University of Toronto, and served as an Assistant Professor in the Department of Psychiatry, Faculty of Medicine, University of Toronto. She is currently an Assistant Teaching Professor in the Master of Professional Studies in Analytics program at Northeastern University, Toronto. Her research interests include machine learning, NLP, and public health analytics.

Vivian Clements Edwin received the Ph.D. degree in Statistics from Bharathiar University, India. He is a Data Science Manager at Deloitte, Toronto, ON, Canada, and a Part-time Lecturer at Northeastern University, Toronto. With over 14 years of experience in enterprise analytics and ML, his research interests include deep learning, NLP, statistical modeling, and MLOps.

Kofi Sarpong Adu-Manu received the Ph.D. degree in computer science from the University of Ghana, Legon, Ghana. He is currently an Associate Professor with the Department of Computer Science, University of Ghana, and a Lead Instructor in AI at the Pan-African Doctoral Academy. His research interests include artificial intelligence, machine learning, wireless sensor networks, IoT, and cybersecurity.

Kwabena Adu Agyemang is a Ph.D. student at New Mexico State University, NM, USA. He holds the M.S. degree in Computer Science from New Mexico State University and the B.S. degree from Valley View University, Ghana. His research interests include AI in education, human-computer interaction, and emerging computing paradigms.